

Experiment-14

Aim: How to run Nmap scan.

Download the Nmap Installer

- This can be found for free from the developer's website. It is highly recommended that you download directly from the developer to avoid any potential viruses or fake files. Downloading the Nmap installer includes Zenmap, the graphical interface for Nmap. ~~Installer includes Zenmap~~, the graphical which makes it easy for newcomers to perform scans without having to learn command lines.
- The Zenmap program is available for windows, linux, and Mac OSx. You can find the installation files for all operating systems on the Nmap website.
 - Install Nmap. Run the installer once it is finished downloading you will be asked which components you would like to install. In order to get the full benefit of Nmap, keep all of these checked. Nmap will not install any adware or spyware.
 - Run the "Nmap - zenmap" GUI program. If you left your settings at default during installation, you should be able to see an icon for it on your desktop. If not, look in your start menu. opening Zenmap will start the program.
 - Enter in the target for your scan. The Zenmap program makes scanning a fairly simple process. The first step to running a scan is choosing your target. you can enter a domain (example.com), an IP address (127.0.0.1), a network (192.168.1.0/24), or a combination of those.
 - choose your profile. profiles are preset groupings of modifiers that change what is scanned. The profiles allow you to quickly select different types of scans without having to type in the modifiers on the command line. choose the profile that best fits your needs: [1].

- Intense Scan - A comprehensive scan. Contains operating system (os) detection, version detection, script scanning, traceroute, and has aggressive scan timing. This is considered an intrusive scan.
- Ping scan - This scan simply detects if the targets are online, it does not scan any ports.
- Quick Scan - This is quicker than a regular scan due to aggressive timing and only scanning select ports.
- Regular scan - This is the standard Nmap scan without any modifiers. It will return ping and return open ports on the target.
- Click Scan to start scanning. The active results of the scan will be displayed in the Nmap output tab. The time the scan takes will depend on the scan profile you choose, the physical distance to the target, and the target's n/w configuration.
- Read your results. Once the scan is finished, you'll see the message "Nmap done" at the bottom of the Nmap output tab. You can now check your results, depending on the type of scan you performed. All of the results will be listed in the main Nmap output tab, but you can use the other tabs to get a better look at specific data.
- Ports/Hosts - This tab will show the results of your port scan, including the services ~~for~~ those ports.
- Topology - This shows the traceroute for the scans you performed.
- Host Details - This shows a summary of your target gained through scans.
- Scans - This tab stores the commands of your previously-run scans.

Experiment-15

Aim : operating system detection using Nmap.

Program

nmap -O -sV x.x.x.x

Starting Nmap 9.92 (<https://nmap.org>) at 2021-09-17 11:02 PDT

Nmap scan report for x.x.x.x

Host is up (0.22s latency).

Not shown: 994 closed tcp ports (reset)

PORT STATE SERVICE VERSION

135/tcp open msrpc Microsoft Windows RPC

139/tcp open netbios-ssn Microsoft Windows netbios-ssn

445/tcp open microsoft-ds?

2201/tcp open cmrcservice Microsoft Configuration Manager Remote Control Service.
(CMRC Service - etc)

3306/tcp open mysql MySQL 5.1.60-community-Leg

3389/tcp open ms-wbt-server Microsoft Terminal Services.

Aggressive OS guesses: Microsoft Windows Server 2016 (93%),

Microsoft Windows Server 2008 R2 SP1 (92%), Microsoft Windows Server 2012 R2 (91%),

Microsoft Windows 10 1607 (90%), Microsoft Windows Server 2008 R2 (90%),

Microsoft Windows Server 2012 (89%), Microsoft Windows Server 2012 or

Windows Server 2012 R2 (89%), Microsoft Windows Server 2008 R2 or Windows 8 (87%),

Microsoft Windows Server 2008 R2 SP1 (86%), Windows 8 (84%), Microsoft Windows

7 SP1 (86%)

No exact OS matches for host (test conditions non-ideal).

Network Distance: 18 hops

Service Info: OS: windows; CPE: cpe:/o:microsoft; windows

OS and service detection performed. Please report any incorrect results at

<https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 83.77 seconds.

Experiment-15

Aim: Operating System Detection using Nmap.

Program

nmap -O -sV x.x.x.x

Starting Nmap 2.9.2 (<https://nmap.org>) at 2021-09-17 11:02 PDT

Nmap scan report for x.x.x.x

Host is up (0.22s latency).

Not shown: 994 closed tcp ports (reset)

PORT STATE SERVICE VERSION

135/tcp open msrpc Microsoft Windows RPC

139/tcp open netbios-ssn Microsoft Windows netbios-ssn

445/tcp open microsoft-ds?

2701/tcp open csmc service Microsoft Configuration Manager Remote Control Service. (CMRC service.cie)

3306/tcp open mysql MySQL 5.1.60-community-Leg

3389/tcp open ms-wbt-server Microsoft Terminal Services.

Aggressive OS guesses: Microsoft Windows Server 2016 (93%),

Microsoft Windows Server 2008 R2 SP1 (92%), Microsoft Windows Server 2012 R2 (91%),

Microsoft Windows 10 1607 (90%), Microsoft Windows Server 2008 R2 (90%),

Microsoft Windows Server 2012 (89%), Microsoft Windows Server 2012 or

Windows Server 2012 R2 (89%), Microsoft Windows Server 2008 R2 or Windows 8 (87%),

Microsoft Windows Server 2008 R2 SP2 (or) Windows 8 (87%), Microsoft Windows

Server 2012 R2 (86%)

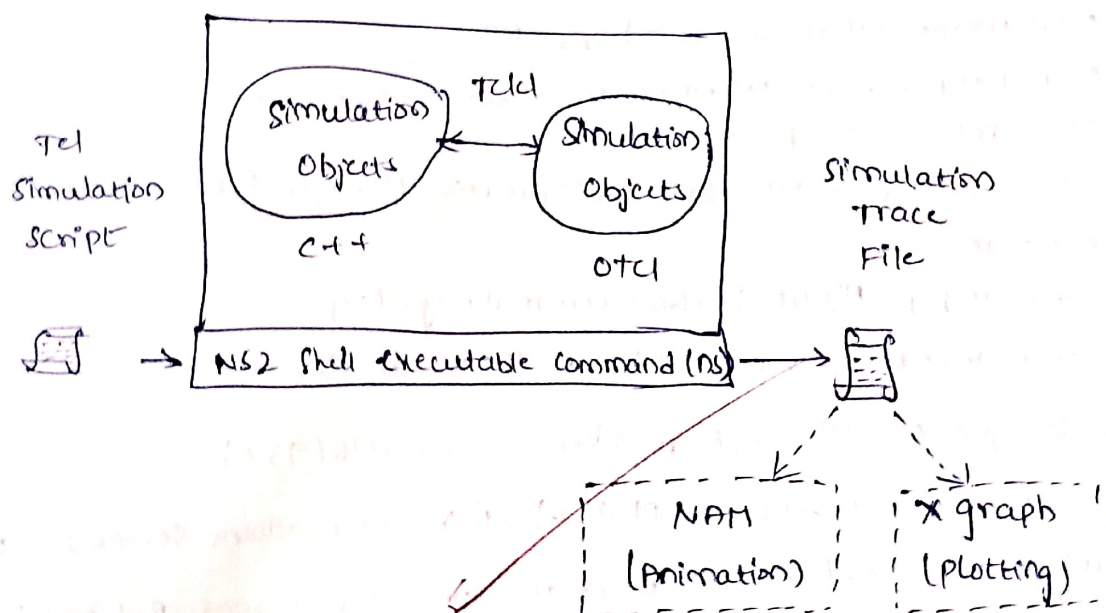
No exact OS matches for host (test conditions non-ideal).

Network Distance: 18 hops

Service Info: OS: windows; CPE: cpe:/o:microsoft; windows

OS and service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 33.77 seconds.



Experiment - 16

Aim : Do the following using NS2 simulator.

(i) NS2 Simulator - Introduction

What is NS2

- 1) NS2 stands for network simulator version
- 2) It is an open-source event-driven simulator designed specifically for research in computer communication n/w's.

Features of NS2

- 1) It is a discrete event simulator for networking research.
- 2) It provides substantial support to simulate bunch of protocols like TCP, FTP, UDP, https and DSR.
- 3) It simulates wired and wireless network.
- 4) It is primarily unix based
- 5) Uses TCL as its scripting language.
- 6) OTC : object oriented support
- 7) TCLK : C++ and OTC linkage.
- 8) Discrete event scheduler.

Basic Architecture

NS2 consists of two key languages: C++ and object-oriented tool command language. While the C++ defines the internal mechanisms of the simulation objects, the OTC sets up simulation by assembling and configuring the objects as well as scheduling discrete events. The C++ and the OTC are linked together using TCLK.